

## NORTHEAST PHILADELPHIA, PA, USA

WMO: 724085

Lat: 40.079N

Lon: 75.013W

Elev: 105

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94732

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 12.8       | 17.2 | -4.8                          | 4.3 | 15.4 | -0.3 | 5.4 | 20.2 | 26.7                  | 32.2 | 24.5 | 31.5 | 10.1                  | 290  | 0.481 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.6                   | 93.3            | 75.0 | 90.8 | 73.8 | 88.4 | 72.6 | 78.0                | 88.6 | 76.5 | 86.2 | 75.3 | 84.0 | 9.6                  | 280  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 75.0                            | 131.8 | 82.8 | 73.4 | 124.7 | 80.7 | 72.5 | 120.7 | 80.2 | 41.4          | 88.7 | 39.9 | 86.7 | 38.7 | 84.4 |                   |
|                                 |       |      |      |       |      |      |       |      |               |      |      |      |      |      |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 22.6              | 19.4 | 17.5 | DB                         | 7.6 | 97.7               | 4.9 | 3.3                                                | 4.0 | 100.0      | 1.2  | 101.9      | -1.6 | 103.8      | -5.2 | 106.2 |
|                   |      |      | WB                         | 5.7 | 81.0               | 4.7 | 3.0                                                | 2.3 | 83.1       | -0.5 | 84.8       | -3.2 | 86.5       | -6.6 | 88.6  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 55.8   | 33.1  | 35.7  | 43.0  | 54.1  | 63.6  | 72.8  | 78.0  | 76.2  | 69.3  | 57.5  | 46.8  | 38.6  |
|                                                                                  | DBStd       | 17.44  | 10.11 | 9.25  | 8.97  | 8.45  | 7.63  | 6.55  | 5.20  | 5.21  | 6.68  | 7.95  | 8.23  | 8.91  |
|                                                                                  | HDD50       | 1787   | 531   | 408   | 254   | 47    | 1     | 0     | 0     | 0     | 18    | 157   | 371   |       |
|                                                                                  | HDD65       | 4628   | 989   | 820   | 682   | 344   | 122   | 12    | 0     | 1     | 32    | 259   | 548   | 819   |
|                                                                                  | CDD50       | 3915   | 7     | 9     | 38    | 170   | 421   | 684   | 867   | 812   | 579   | 251   | 60    | 17    |
|                                                                                  | CDD65       | 1279   | 0     | 0     | 1     | 16    | 78    | 246   | 403   | 347   | 161   | 26    | 1     | 0     |
|                                                                                  | CDH74       | 11971  | 0     | 1     | 13    | 184   | 763   | 2300  | 4132  | 3180  | 1219  | 177   | 2     | 0     |
|                                                                                  | CDH80       | 4446   | 0     | 0     | 1     | 46    | 231   | 834   | 1759  | 1185  | 360   | 30    | 0     | 0     |
| Wind                                                                             | WSAvg       | 7.8    | 8.8   | 9.0   | 9.5   | 9.0   | 7.7   | 7.2   | 6.6   | 6.1   | 6.7   | 7.2   | 7.8   | 8.2   |
| Precipitation                                                                    | PrecAvg     | 46.5   | 3.7   | 2.7   | 4.2   | 3.9   | 3.5   | 4.1   | 4.4   | 4.8   | 4.0   | 3.8   | 3.3   | 4.1   |
|                                                                                  | PrecMax     | 62.1   | 5.9   | 5.4   | 7.5   | 10.3  | 7.4   | 9.2   | 10.7  | 13.5  | 9.8   | 9.6   | 5.7   | 10.0  |
|                                                                                  | PrecMin     | 36.8   | 1.6   | 0.7   | 1.3   | 1.2   | 0.9   | 1.1   | 1.2   | 1.4   | 0.5   | 0.6   | 1.1   | 0.8   |
|                                                                                  | PrecStd     | 6.5    | 1.3   | 1.2   | 1.6   | 1.9   | 1.4   | 2.1   | 2.0   | 2.5   | 2.4   | 2.2   | 1.3   | 1.9   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 65.3  | 67.9  | 76.1  | 85.6  | 89.8  | 94.2  | 97.7  | 95.9  | 92.3  | 83.4  | 72.9  | 68.5  |
|                                                                                  |             | MCWB   | 60.2  | 58.9  | 60.8  | 64.5  | 71.6  | 75.1  | 77.4  | 76.9  | 73.8  | 70.3  | 63.3  | 61.5  |
|                                                                                  | 2%          | DB     | 60.1  | 61.0  | 68.6  | 79.5  | 86.1  | 90.8  | 94.1  | 91.8  | 87.7  | 79.0  | 68.5  | 62.6  |
|                                                                                  |             | MCWB   | 55.3  | 53.0  | 56.6  | 62.9  | 69.6  | 73.6  | 75.4  | 74.5  | 71.1  | 67.6  | 60.0  | 57.2  |
|                                                                                  | 5%          | DB     | 53.2  | 54.8  | 63.3  | 74.1  | 82.0  | 88.0  | 91.3  | 89.3  | 84.0  | 74.8  | 64.3  | 57.1  |
|                                                                                  |             | MCWB   | 48.2  | 47.0  | 53.2  | 60.0  | 68.0  | 72.6  | 74.5  | 73.2  | 69.8  | 65.2  | 57.4  | 52.2  |
|                                                                                  | 10%         | DB     | 47.6  | 49.6  | 58.0  | 69.6  | 78.4  | 84.7  | 88.8  | 86.4  | 81.1  | 71.2  | 60.8  | 52.2  |
|                                                                                  |             | MCWB   | 43.0  | 43.5  | 49.5  | 57.8  | 65.5  | 70.4  | 73.2  | 71.8  | 68.8  | 63.1  | 54.4  | 46.9  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 62.0  | 61.4  | 63.9  | 68.6  | 74.3  | 78.2  | 81.6  | 80.3  | 77.3  | 73.4  | 65.8  | 63.4  |
|                                                                                  |             | MCDB   | 64.2  | 68.1  | 73.5  | 78.7  | 85.5  | 88.0  | 91.1  | 90.3  | 87.0  | 78.6  | 70.3  | 66.2  |
|                                                                                  | 2%          | WB     | 56.0  | 54.7  | 58.8  | 65.4  | 71.8  | 76.0  | 78.3  | 77.6  | 75.1  | 70.2  | 62.6  | 58.1  |
|                                                                                  |             | MCDB   | 59.6  | 59.3  | 66.1  | 75.5  | 82.6  | 86.9  | 89.6  | 87.5  | 81.9  | 76.6  | 66.4  | 61.7  |
|                                                                                  | 5%          | WB     | 48.5  | 48.4  | 54.7  | 62.5  | 69.6  | 74.7  | 76.8  | 76.0  | 73.3  | 67.3  | 59.0  | 53.3  |
|                                                                                  |             | MCDB   | 51.6  | 53.2  | 61.2  | 71.5  | 79.1  | 84.6  | 87.1  | 84.6  | 79.6  | 73.0  | 63.2  | 56.9  |
|                                                                                  | 10%         | WB     | 43.8  | 43.8  | 50.0  | 59.3  | 67.4  | 73.0  | 75.4  | 74.6  | 71.3  | 64.1  | 55.4  | 47.7  |
|                                                                                  |             | MCDB   | 47.2  | 49.0  | 57.7  | 68.1  | 75.8  | 81.8  | 84.5  | 82.3  | 78.2  | 69.2  | 59.8  | 51.2  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 14.2  | 15.6  | 17.5  | 19.7  | 18.8  | 18.1  | 17.6  | 17.2  | 17.8  | 17.8  | 16.8  | 14.1  |
|                                                                                  |             | MCDBR  | 21.5  | 23.4  | 25.8  | 26.2  | 23.3  | 20.6  | 20.1  | 19.2  | 20.6  | 21.4  | 21.3  | 20.0  |
|                                                                                  |             | MCWBR  | 18.1  | 17.7  | 16.9  | 13.9  | 11.2  | 9.0   | 7.5   | 7.6   | 8.8   | 12.1  | 15.0  | 17.5  |
|                                                                                  | 5% WB       | MCDBR  | 20.0  | 21.8  | 23.9  | 22.8  | 21.0  | 18.4  | 17.7  | 16.1  | 16.1  | 17.1  | 17.6  | 19.0  |
|                                                                                  |             | MCWBR  | 18.3  | 18.5  | 18.0  | 14.8  | 11.5  | 9.7   | 8.3   | 7.4   | 9.0   | 11.7  | 15.0  | 18.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.312 | 0.321 | 0.361 | 0.392 | 0.439 | 0.501 | 0.501 | 0.494 | 0.407 | 0.361 | 0.328 | 0.310 |
|                                                                                  | taud        |        | 2.456 | 2.419 | 2.372 | 2.313 | 2.212 | 2.080 | 2.095 | 2.101 | 2.315 | 2.454 | 2.508 | 2.515 |
|                                                                                  | Ebn at Noon |        | 271   | 284   | 283   | 280   | 268   | 251   | 249   | 247   | 264   | 266   | 263   | 263   |
|                                                                                  | Edh at Noon |        | 25    | 30    | 35    | 40    | 45    | 51    | 50    | 49    | 37    | 29    | 24    | 22    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 607   | 895   | 1183  | 1531  | 1702  | 1865  | 1903  | 1650  | 1370  | 974   | 691   | 526   |
|                                                                                  | RadStd      |        | 53    | 98    | 95    | 135   | 199   | 154   | 91    | 106   | 150   | 101   | 62    | 53    |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            |       | +1.12   | N/A    | N/A     | N/A   | N/A   | N/A         | -232  | +342  | +214  |
| (47) Regional (72 neighbors) |       | +0.72   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +220  | +124  |

Nomenclature: See separate page